darkniche - Reference Manual

August 7, 2026

Title Climatic Dark Diversity and Directional Niche Coverage

Version 0.1.0

Description A reproducible framework for asking which species are absent from a region despite tolerating its climate -- its climatic dark diversity. Implements the directional niche-coverage family (DNC_q, DNC_auc, NSI, CFI, NOV), which answers containment questions that symmetric overlap statistics (Schoener's D, Warren's I) cannot express; three analytical null models (breadth-matched, environmental shift, geographic relocation); calibrated dark-diversity accounting with an area-adjusted dark-diversity ratio; validation against documented introduced ranges with fast DeLong tests; and a threshold-sweep diagnostic that reveals when the direction of an aggregate null-model test is an artefact of the chosen threshold. Publication-quality figures and table exporters are included, together with results for all 619 terrestrial ecoregions as bundled example data.

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Encoding UTF-8

Roxygen list(markdown = TRUE)

RoxygenNote 7.3.1

Depends R (>= 4.1)

Imports stats, graphics, grDevices, utils, tools

Suggests testthat (>= 3.0.0), knitr, rmarkdown, terra, pkgdown

Config/testthat/edition 3

VignetteBuilder knitr

URL https://github.com/ehsanrahimi666/darkniche, https://ehsanrahimi666.github.io/darkniche/

BugReports https://github.com/ehsanrahimi666/darkniche/issues

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`dn_aggregate` *Region-level aggregate admissibility test (single threshold)*

Description

Tests whether a region holds more climatically admissible species than the null expects: the observed count of species whose coverage exceeds `threshold` against the Poisson-binomial expectation of the same count under each species' own null. One rule, both sides – applying a stricter criterion to the observed side than the null side guarantees a negative result regardless of the data.

Usage

```
dn_aggregate(obs, null_mean, null_sd, threshold)
```

Arguments

obs	length-n observed coverage values (one per species).
null_mean, null_sd	length-n per-species null moments.
threshold	coverage threshold.

Details

Only nulls whose marginals are genuinely counterfactual support this test (the environmental-shift null N2 or relocation null N3); the breadth-matched null reproduces the observed marginal by construction and has no power here.

Results with expected count < 5 are flagged `reliable = FALSE`: the normal approximation to the Poisson-binomial count is invalid there and the SES/enrichment are arithmetic, not evidence. **Never report unreliable cells.**

Value

one-row data.frame: `n_species`, `n_admissible_obs`, `expected`, `sd`, `dn_ses`, `p`, `enrichment`, `reliable`.

Examples

```
set.seed(1)
obs <- c(rbeta(300, 0.3, 3), rbeta(100, 4, 1.5))   # bimodal coverage
mu <- rbeta(400, 1, 3); sdv <- 0.05 + 0.1 * runif(400)
dn_aggregate(obs, mu, sdv, threshold = quantile(obs, 0.95))
```

---

dn_auc *AUC with DeLong variance (fast algorithm of Sun & Xu 2014)*

Description

AUC with DeLong variance (fast algorithm of Sun & Xu 2014)

Usage

```
dn_auc(score, label)
```

Arguments

score	numeric vector; higher = more positive.
label	logical or 0/1 vector.

Value

list with auc, var, ci (95%).

Examples

```
set.seed(1)
lab <- rep(c(TRUE, FALSE), each = 200)
score <- ifelse(lab, rnorm(200, 1), rnorm(200, 0))
dn_auc(score, lab)
```

---

dn_calibrate *Calibrate the admissibility threshold (Youden)*

Description

Calibrate the admissibility threshold (Youden)

Usage

```
dn_calibrate(score, label)
```

Arguments

score	establishment-model probabilities (or any coverage score).
label	validation labels.

Value

the Youden-optimal threshold.

Examples

```
set.seed(1)
lab <- rep(c(TRUE, FALSE), each = 200)
p <- ifelse(lab, rbeta(200, 4, 2), rbeta(200, 2, 4))
dn_calibrate(p, lab)     # Youden-optimal threshold
```

dn_cddr *Climatic dark-diversity ratio and its area adjustment*

Description

`CDDR = n_dark / (n_dark + n_present)`. The raw ratio is confounded with region area (a small region occupies little environmental space, so more niches enclose it, while its smaller area supports fewer natives; in the global analysis r = -0.44), so cross-region comparison must use the residual from the CDDR ~ log(area) regression. **Never rank regions by raw CDDR.**

Usage

```
dn_cddr(n_dark, n_present, area = NULL)
```

Arguments

`n_dark, n_present`	integer vectors per region.
`area`	region areas (same order); required for `adjusted`.

Value

data.frame with `dn_cddr`, and when `area` is given `cddr_residual` plus the fitted confound as attributes `fit`.

Examples

```
set.seed(1)
area <- 10^runif(50, 4, 6.7)
nd <- rpois(50, 40); np <- rpois(50, 60)
x <- dn_cddr(nd, np, area)
attr(x, "confound_r")     # never rank regions by raw cddr
head(x)
```

dn_centres *Lattice cell centres*

Description

Lattice cell centres

Usage

```
dn_centres(sp)
```

Arguments

sp	a dn_space

Value

list with x, y centre vectors

Examples

```
sp <- dn_space(cbind(rnorm(1000), rnorm(1000)), R = 40)
ctr <- dn_centres(sp)
str(ctr)
```

---

dn_dark *Dark-diversity classification of species-region pairs*

Description

Applies the full conjunction rule of the source studies: a pair is *climatically admissible* if the calibrated establishment probability exceeds p_min and every supplied null screen has SES >= ses_min; it is *dark* if additionally the species is absent from the region.

Usage

```
dn_dark(pairs, p_min = 0.177, ses_cols = character(), ses_min = 2)
```

Arguments

pairs	data.frame with at least species, region, present (logical), a probability column p_establish, and one or more SES columns given in ses_cols.
p_min	admissibility threshold on the establishment probability. The source studies' externally calibrated value (Youden threshold of a species-blocked, climate-only model validated on documented introduced ranges) is 0.177; recalibrate with dn_calibrate() when using a different model or metric.
ses_cols	character vector of SES column names (e.g. c("n1_ses", "n2_ses", "n3_ses")).
ses_min	SES criterion (default 2).

Value

pairs with logical columns admissible and dark.

Examples

```
pairs <- data.frame(species = c("a", "a", "b"), region = c("r1", "r2", "r1"),
                    present = c(TRUE, FALSE, FALSE),
                    p_establish = c(0.6, 0.4, 0.05),
                    n2_ses = c(3.1, 2.5, 4.0))
dn_dark(pairs, ses_cols = "n2_ses")
```

dn_data *Load bundled ecoregion example data*

Description

Results of the global analysis of Rahimi & Jung (in review) for all 619 WWF terrestrial ecoregions (35,087 species; 21,718,853 species-ecoregion pairs), shipped so that every example, test and vignette in `darkniche` runs on real output rather than on toy numbers.

Usage

```
dn_data(
  which = c("ecoregion", "sweep", "sources", "taxon", "region_taxon", "auc",
    "containment")
)
```

Arguments

`which`	name of the dataset (see above).

Details

Available datasets:

`ecoregion` 619 rows: `region`, `n_obs`, `n_dark`, `cddr`, `area_km2`.

`sweep` 2,476 rows: `region`, `quantile`, `threshold`, `enrichment`, `reliable` – the aggregate null test at four thresholds per ecoregion.

`sources` 619 rows: `near`, `regional`, `distant` shares of each ecoregion's dark diversity by native-range distance.

`taxon` dark-diversity rate per taxonomic group with 95% CI.

`region_taxon` 3,095 rows: CDDR for each ecoregion x taxon.

`auc` AUC of each metric against documented introduced ranges.

`containment` AUC by niche-to-region size stratum.

Value

a data.frame.

Examples

```
eco <- dn_data("ecoregion")
str(eco)
summary(eco$cddr)

# metric performance against documented introductions
dn_data("auc")
```

dn_delong *Paired DeLong test between two correlated AUCs*

Description

Paired DeLong test between two correlated AUCs

Usage

```
dn_delong(s1, s2, label)
```

Arguments

s1, s2	two score vectors on the same cases.
label	logical or 0/1 vector.

Value

list with auc1, auc2, z, p (two-sided).

Examples

```
set.seed(1)
lab <- rep(c(TRUE, FALSE), each = 300)
s1 <- ifelse(lab, rnorm(300, 1.0), rnorm(300, 0))
s2 <- s1 + rnorm(600, 0, 0.8)      # a noisier version of the same signal
dn_delong(s1, s2, lab)
```

dn_density *Kernel density of a point set on the lattice*

Description

Bins the points on the lattice (half-open bins, right edge of the last bin closed, exactly as numpy.histogram2d), smooths with a Gaussian kernel of sp$bw_cells cells, and normalises to unit mass.

Usage

```
dn_density(points, sp, weights = NULL)
```

Arguments

points	numeric matrix (n x 2) of scores (e.g. a species' occupied cells projected into the space, or a region's cells).
sp	a dn_space.
weights	optional per-point weights (e.g. cell areas).

Value

numeric `R x R` matrix summing to 1 (all-zero if no point falls inside the lattice).

Examples

```
set.seed(1)
sp <- dn_space(cbind(rnorm(5000), rnorm(5000)), R = 60)
pts <- cbind(rnorm(300, 0.5, 0.4), rnorm(300, -0.2, 0.5))
Z <- dn_density(pts, sp)
sum(Z)                                  # unit mass
image(Z, main = "species niche density")
```

---

dn_direction_census *Direction census of an aggregate null test across thresholds*

Description

Summarises, over many regions, how many are called enriched and how many depleted at each threshold, and what share of regions change direction somewhere in the sweep. This is the quantity a study must report before any single-threshold enrichment claim can be trusted.

Usage

```
dn_direction_census(sweep, reliable_only = TRUE)
```

Arguments

`sweep`	a long data.frame with columns `region`, `quantile` and `enrichment` (observed/expected), and optionally `reliable`.
`reliable_only`	if `TRUE` (default) only rows flagged reliable are used; unreliable cells (expected count < 5) are not interpretable.

Value

list with `census` (per threshold: n enriched, n depleted), `n_regions`, `n_changed`, `frac_changed`, `n_monotone`.

Examples

```
cen <- dn_direction_census(dn_data("sweep"))
cen$census
round(cen$frac_changed, 3)     # share of ecoregions that flip direction
```

dn_entryq *Entry quantile of every lattice cell*

Description

For each cell, the smallest envelope level q at which it enters the q-envelope – its cumulative density rank within the niche. Cells at the mode enter near 0; peripheral cells enter near 1; unoccupied cells are assigned 1. `1 - dn_entryq(Z)` is the weight vector whose inner product with a region density gives `DNC_auc`.

Usage

```
dn_entryq(Z)
```

Arguments

`Z`	density vector (or matrix, flattened internally).

Value

numeric vector of entry quantiles in [0, 1].

Examples

```
sp <- dn_space(cbind(rnorm(3000), rnorm(3000)), R = 50)
Z <- dn_density(cbind(rnorm(200), rnorm(200)), sp)
q <- dn_entryq(Z)
range(q)                                  # mode enters near 0, edges near 1
```

dn_envelope *q-quantile niche envelope*

Description

The smallest set of lattice cells containing a fraction `q` of the density mass. This replaces the naive `Z > 0` support, which after kernel smoothing covers most of the lattice and makes coverage metrics bandwidth-driven rather than ecology-driven.

Usage

```
dn_envelope(Z, q = 0.95, flat = FALSE)
```

Arguments

`Z`	density as a matrix or vector (one object), or a matrix with one *row per object* if `flat = TRUE`.
`q`	envelope level in (0, 1]; `q >= 1` returns the positive support.
`flat`	if `TRUE`, treat `Z` as (m x P) flattened densities and return an (m x P) logical matrix.

Value

logical vector/matrix marking envelope membership.

Examples

```
sp <- dn_space(cbind(rnorm(3000), rnorm(3000)), R = 50)
Z <- dn_density(cbind(rnorm(200), rnorm(200)), sp)
E <- dn_envelope(Z, q = 0.95)
mean(E)                                  # share of lattice in the envelope
```

---

dn_match *Realm- and area-matched negatives for introduced-range validation*

Description

For each positive (region a species actually reached), samples `ratio` negatives from unreached regions in the same biogeographic realm with the nearest log-areas – matching without which negatives are dominated by climatically implausible regions and every metric rejects them equally.

Usage

```
dn_match(positives, regions, reached, ratio = 5L, seed = 1L)
```

Arguments

`positives`	data.frame with `species`, `region`.
`regions`	data.frame with `region`, `realm`, `area`.
`reached`	list mapping species to reached regions.
`ratio`	negatives per positive (default 5).
`seed`	RNG seed.

Value

data.frame `species`, `region`, `label`.

Examples

```
regs <- data.frame(region = paste0("r", 1:8), realm = rep(c("PA", "NT"), 4),
                   area = 10^runif(8, 4, 6))
pos <- data.frame(species = "sp1", region = "r1")
dn_match(pos, regs, reached = list(sp1 = "r1"), ratio = 3)
```

dn_metrics *All pairwise niche metrics for a block of species and regions*

Description

Computes, for every (species, region) pair, the symmetric baselines (Schoener's D, Warren's I, the Bhattacharyya coefficient), the directional coverage family (DNC_q, its threshold-free integral DNC_auc, the converse share NSI_q), the min-overlap identities (DNC_min, NSI_min; with unit-mass densities DNC_min == D, the algebraic result that no symmetric functional of the min family can express containment), the novelty share NOV, and – when moments are supplied – the Mahalanobis core-fit index CFI and its availability-weighted form CFI_w.

Usage

```
dn_metrics(
  Zs,
  Ze,
  q = 0.95,
  sp_mu = NULL,
  sp_cov = NULL,
  eco_mu = NULL,
  centres = NULL
)
```

Arguments

Zs	(m x P) matrix: one flattened species density per row.
Ze	(k x P) matrix: one flattened region density per row.
q	envelope level for the threshold form (default 0.95).
sp_mu, sp_cov	optional species moments: (m x 2) means and a list of m 2x2 covariance matrices (or an (m x 4) matrix, row-major).
eco_mu	optional (k x 2) region centroids.
centres	optional dn_centres(sp) list, required for CFI_w.

Value

named list of (m x k) matrices.

Examples

```
set.seed(1)
sp <- dn_space(cbind(rnorm(5000, 0, 2), rnorm(5000, 0, 2)), R = 60)
# broad species niche engulfing a small region: the containment case
broad  <- dn_density(cbind(rnorm(1500, 0, 1.5), rnorm(1500, 0, 1.5)), sp)
region <- dn_density(cbind(rnorm(400, 0, 0.25), rnorm(400, 0, 0.25)), sp)
m <- dn_metrics(matrix(as.numeric(broad), 1), matrix(as.numeric(region), 1))
# D is near zero although the region is trivially habitable:
round(c(D = m$D, DNC_q = m$DNC_q, DNC_auc = m$DNC_auc), 3)
```

dn_moments *First and second moments of a lattice density*

Description

Mean vector and covariance matrix of a density in environmental space, used by the core-fit index (CFI).

Usage

```
dn_moments(Z, sp)
```

Arguments

Z	R x R density matrix.
sp	a dn_space.

Value

list with mu (length 2) and cov (2 x 2).

Examples

```
sp <- dn_space(cbind(rnorm(3000), rnorm(3000)), R = 50)
Z <- dn_density(cbind(rnorm(200, 1, 0.3), rnorm(200, 0, 0.4)), sp)
dn_moments(Z, sp)
```

dn_null_breadth *Breadth-matched null (N1)*

Description

Standardises an (n_species x n_region) metric matrix against reference species of similar niche breadth *within taxon*: for each species, the K nearest neighbours in log10 breadth form the reference set. This null asks whether a given species is exceptional relative to species of its own niche breadth. It cannot support the region-level aggregate test (its marginal reproduces the observed distribution by construction); use the shift or relocation null there.

Usage

```
dn_null_breadth(values, breadth, taxon, K = 200L, rel_floor = 0.02)
```

Arguments

values	(n x k) metric matrix.
breadth	length-n niche breadths (any monotone measure).
taxon	length-n taxon labels.
K	reference window size (default 200).
rel_floor	SES floor, see dn_ses().

Value

list with dn_ses, null_mean, null_sd, p_emp (empirical permutation p with floor 1/(n_ref+1)), each (n x k).

Examples

```
set.seed(1)
V <- matrix(runif(30 * 3), 30, 3)          # 30 species x 3 regions
br <- rlnorm(30)
nb <- dn_null_breadth(V, br, taxon = rep("bird", 30), K = 10)
head(nb$ses)
```

---

dn_null_relocate *Geographic relocation null (N3)*

Description

Null coverage obtained by stamping the species' occupied-cell footprint at random positions on the geographic grid and recomputing coverage from the relocated range. Requires the geographic cell adjacency of the analysis grid; supply project to map relocated geographic cells to environmental scores.

Usage

```
dn_null_relocate(
  cells,
  grid_dim,
  valid,
  project,
  sp,
  Ze,
  q = 0.95,
  statistic = c("auc", "q"),
  B = 199L,
  seed = 1L
)
```

Arguments

`cells`	integer matrix (n x 2) of occupied geographic cell indices.
`grid_dim`	geographic grid dimensions `c(nrow, ncol)`.
`valid`	logical matrix marking terrestrial cells.
`project`	function taking an (n x 2) cell-index matrix and returning (n x 2) environmental scores (NA rows for cells without climate).
`sp`	a `dn_space`.
`Ze`	(k x P) region densities.
`q, statistic, B, seed`	as in `dn_null_shift()`.

Value

(k x B) matrix of null coverage values (NA where a relocation could not be placed).

Examples

```
# Requires a geographic grid and a projection function; sketch:
## Not run:
nul <- dn_null_relocate(cells, grid_dim = dim(valid), valid = valid,
                        project = my_lookup, sp = sp, Ze = Ze, B = 99)

## End(Not run)
```

`dn_null_shift` *Environmental-shift null (N2)*

Description

Null distribution of directional coverage obtained by translating the species' coverage-weight array to random, availability-weighted positions in environmental space. Translation permutes lattice cells without altering density values, so the envelope of a shifted niche is exactly the shifted envelope; the weights are computed once and rolled.

Usage

```
dn_null_shift(
  Z_sp,
  Ze,
  Z_global,
  q = 0.95,
  B = 499L,
  statistic = c("auc", "q"),
  targets = NULL,
  seed = NULL
)
```

Arguments

Z_sp	R x R species density.
Ze	(k x P) region densities.
Z_global	R x R global availability density; shift targets are drawn proportional to it.
q	envelope level.
B	number of replicates (default 499).
statistic	"q" or "auc".
targets	optional integer vector of length B giving flat target cell indices (1-based, column-major on the R x R lattice); supply to reproduce a specific draw exactly (e.g. from the Python pipeline).
seed	RNG seed used when targets is NULL.

Value

(k x B) matrix of null coverage values.

Examples

```
set.seed(1)
bg <- cbind(rnorm(4000), rnorm(4000))
sp <- dn_space(bg, R = 50)
Zg <- dn_density(bg, sp)
Zs <- dn_density(cbind(rnorm(300, 1, 0.4), rnorm(300, 0, 0.4)), sp)
Ze <- matrix(as.numeric(dn_density(cbind(rnorm(300, 1, 0.3),
                                         rnorm(300, 0, 0.3)), sp)), 1)
obs <- as.numeric(Ze %*% dn_weights(Zs))
nul <- dn_null_shift(Zs, Ze, Zg, B = 99, seed = 1)
dn_null_summary(obs, nul)
```

---

dn_null_summary *Summarise a null matrix*

Description

Summarise a null matrix

Usage

```
dn_null_summary(obs, null, scale = NULL, rel_floor = 0.02)
```

Arguments

obs	length-k observed values.
null	(k x B) null matrix.
scale, rel_floor	SES flooring, see dn_ses().

Value

data.frame with null mean, sd, SES, empirical p, n valid.

Examples

```
obs <- c(0.5, 0.2)
nul <- matrix(runif(2 * 99, 0, 0.4), 2, 99)
dn_null_summary(obs, nul)
```

`dn_plot_cddr_area` *CDDR against area with the confound displayed (papers' Figure 7 style)*

Description

Raw ratio against log area with the fitted confound line, the top positive and negative residuals labelled – the display that shows the confound rather than hiding it.

Usage

```
dn_plot_cddr_area(cddr_tab, area, labels = NULL, n_label = 5)
```

Arguments

`cddr_tab`	output of `dn_cddr()` (with `area` supplied).
`area`	the area vector used.
`labels`	optional region names for labelling extremes.
`n_label`	how many extremes to label on each side (default 5).

Value

invisibly, the labelled subset.

Examples

```
set.seed(1)
area <- 10^runif(80, 4, 6.7)
nd <- rpois(80, 30 + 200 / log10(area)); np <- rpois(80, 60)
x <- dn_cddr(nd, np, area)
dn_plot_cddr_area(x, area, labels = paste0("eco", 1:80))
```

dn_plot_containment *Conceptual containment figure (papers' Figure 1)*

Description

Generates the three-panel demonstration that symmetric overlap cannot answer a containment question: a region nested in a niche core, nested at the margin, and outside the niche, with D, DNC_q and DNC_auc annotated.

Usage

```
dn_plot_containment(R = 100, broad = 14, narrow = 4, offsets = c(0, 18, 34))
```

Arguments

R	lattice resolution (default 100).
broad, narrow	Gaussian widths (lattice cells) of niche and region.
offsets	horizontal offsets of the region for the three panels.

Value

invisibly, a data.frame of the annotated metric values.

Examples

```
dn_plot_containment()
```

dn_plot_density *Plot a density in environmental space*

Description

Filled image of a lattice density with optional envelope contour and a second density overlaid as contour lines – the building block of species-vs-region panels.

Usage

```
dn_plot_density(Z, sp, q = 0.95, overlay = NULL, main = "", ...)
```

Arguments

Z	R x R density (species or region).
sp	a dn_space.
q	optional envelope level to outline (default 0.95; NA for none).
overlay	optional second density drawn as contours (e.g. the region over a species niche).
main	title.
...	passed to graphics::image().

Value

invisibly, Z.

Examples

```
set.seed(1)
sp <- dn_space(cbind(rnorm(4000), rnorm(4000)), R = 60)
Zs <- dn_density(cbind(rnorm(800, 0, 1), rnorm(800, 0, 1)), sp)
Ze <- dn_density(cbind(rnorm(300, 0.4, 0.25), rnorm(300, 0, 0.25)), sp)
dn_plot_density(Zs, sp, overlay = Ze, main = "region inside niche")
```

---

dn_plot_matrix *Region-by-taxon matrix heatmap (papers' Figure 9 style)*

Description

Region-by-taxon matrix heatmap (papers' Figure 9 style)

Usage

```
dn_plot_matrix(M, main = "", values = TRUE, key = TRUE)
```

Arguments

M	numeric matrix, regions in rows, taxa in columns (e.g. CDDR).
main	title.
values	logical, print cell values (default TRUE).
key	logical, draw a colour key in the right margin (default TRUE).

Value

invisibly, M.

Examples

```
M <- matrix(runif(6 * 5), 6, 5,
            dimnames = list(paste("region", 1:6),
                            c("amph", "birds", "mamm", "plants", "rept")))
dn_plot_matrix(M, main = "CDDR by region and taxon")
```

dn_plot_roc *ROC curves for metric comparison (papers' Figure 3 style)*

Description

ROC curves for metric comparison (papers' Figure 3 style)

Usage

```
dn_plot_roc(scores, label, main = "introduced-range validation")
```

Arguments

scores	named list of score vectors on the same cases.
label	logical/0-1 vector.
main	title.

Value

invisibly, data.frame of AUCs with 95% CIs.

Examples

```
set.seed(1)
lab <- rep(c(TRUE, FALSE), each = 300)
s1 <- ifelse(lab, rnorm(300, 1), rnorm(300, 0))
s2 <- s1 + rnorm(600, 0, 1)
dn_plot_roc(list(DNC_q = s1, D = s2), lab)
```

dn_plot_sources *Stacked source-decomposition bars (papers' Figure 11 style)*

Description

Stacked source-decomposition bars (papers' Figure 11 style)

Usage

```
dn_plot_sources(
  shares,
  main = "source of dark diversity",
  legend_pos = "topright",
  cex_names = 0.7
)
```

Arguments

shares	matrix or data.frame with columns near, regional, distant (proportions summing to 1), regions in rows.
main	title.
legend_pos	legend position, or NA to suppress it.
cex_names	label size for the region names.

Value

invisibly, shares.

Examples

```
s <- t(sapply(1:8, function(i) { x <- runif(3); x / sum(x) }))
colnames(s) <- c("near", "regional", "distant")
rownames(s) <- paste("region", 1:8)
dn_plot_sources(s)
```

---

dn_plot_sweeps *Multi-region threshold-sweep figure (papers' Figure 10 style)*

Description

Draws every region's observed/expected curve on a log axis with the cross-region median, colouring curves by their direction at the lowest threshold – the display in which the bimodal-vs-unimodal crossing is visible at a glance.

Usage

```
dn_plot_sweeps(sweeps, main = "aggregate null test across thresholds")
```

Arguments

sweeps	a list of dn_sweep() results (one per region).
main	title.

Value

invisibly, the share of regions whose direction reverses.

Examples

```
set.seed(1)
sweeps <- lapply(1:15, function(i) {
  obs <- c(rbeta(250, 0.3, 3), rbeta(80, 4, 1.5))
  dn_sweep(obs, rbeta(330, 1, 2.2), 0.08 + 0.1 * runif(330))
})
dn_plot_sweeps(sweeps)
```

dn_report *One-call region report*

Description

Assembles the quantities a region-level dark-diversity study must report together: raw ratio, area-adjusted residual, rank on each, and the direction census of the region's own threshold sweep. Printing the result surfaces the two failure modes – an area-driven ranking and a threshold-driven direction – instead of leaving them to a footnote.

Usage

```
dn_report(eco, sweep = NULL)
```

Arguments

eco	data.frame with region, n_obs, n_dark, area_km2.
sweep	optional long sweep table (as dn_data() "sweep").

Value

object of class dn_report.

Examples

```
rep <- dn_report(dn_data("ecoregion"), dn_data("sweep"))
rep
head(rep$table)
```

dn_ses *Standardised effect size with a floored null SD*

Description

Reference distributions for coverage metrics are heavily zero-inflated; when a stratum piles up at zero the null SD collapses and SES values in the hundreds are arithmetic, not evidence. The null SD is floored at rel_floor * scale (2% of the between-species spread by default), which preserves ranking and compresses only the absurd tail.

Usage

```
dn_ses(obs, null_mean, null_sd, scale = NULL, rel_floor = 0.02, min_sd = 1e-09)
```

Arguments

obs, null_mean, null_sd	numeric vectors.
scale	spread of the metric across all species (per region), used for the floor; NULL disables flooring.
rel_floor	floor as a fraction of scale (default 0.02).
min_sd	SDs below this become NA (default 1e-9).

Value

numeric vector of SES values.

Examples

```
obs <- c(0.4, 0.1, 0.8)
dn_ses(obs, null_mean = c(0.1, 0.1, 0.1), null_sd = c(0.05, 1e-6, 0.2),
       scale = 0.3)   # the 1e-6 SD is floored, not divided by
```

---

dn_space *Define a shared environmental-space lattice*

Description

Builds the common R x R lattice on which all species niches and all region climate availabilities are estimated, from the global distribution of environmental scores (typically the first two principal components of climate over all terrestrial cells).

Usage

```
dn_space(scores, R = 100L, clip_q = c(0.001, 0.999), bw_frac = 0.02)
```

Arguments

scores	numeric matrix (n x 2) of environmental scores for the *global* background (all terrestrial grid cells), not for any one species or region. The lattice must be shared or no two densities are comparable.
R	lattice resolution per axis (default 100, as calibrated in the source studies; sensitivity there: Spearman >= 0.996 across 60-150).
clip_q	lower/upper quantiles at which the axes are clipped before 5% padding (default c(0.001, 0.999)).
bw_frac	fixed kernel bandwidth as a fraction of the axis range (default 0.02; one bandwidth for every object so that estimated niche breadth reflects ecology, not sample size).

Value

an object of class dn_space: list with x_edges, y_edges, R, bw_cells.

Examples

```
set.seed(1)
bg <- cbind(rnorm(5000), rnorm(5000))   # global background scores
sp <- dn_space(bg, R = 60)
sp
```

---

dn_sweep *Threshold-sweep diagnostic for the aggregate null test*

Description

The central methodological caution of the source studies: the direction of the single-threshold aggregate test is, in general, a property of the threshold, not of the region – in the global analysis its verdict reversed with the threshold in 89% of the world's ecoregions, because observed coverage is bimodal while null coverage is unimodal, so the two distributions must cross. Any aggregate null test on a thresholded, bimodally distributed similarity statistic should either demonstrate that the sign of its result is stable across a stated threshold range, or report the full sweep.

Usage

```
dn_sweep(
  obs,
  null_mean,
  null_sd,
  quantiles = seq(0.8, 0.99, by = 0.01),
  pooled_obs = NULL
)
```

Arguments

obs	length-n observed coverage values for one region.
null_mean, null_sd	per-species null moments (as dn_aggregate()).
quantiles	threshold quantiles of the observed distribution (default seq(0.80, 0.99, by = 0.01)).
pooled_obs	optional vector from which threshold quantiles are computed (e.g. all regions pooled); defaults to obs.

Details

Thresholds are placed at quantiles of the pooled observed coverage distribution, so the sweep is comparable across regions.

Value

object of class dn_sweep_result: data.frame with one row per threshold plus attributes direction_change (logical) and monotone.

Examples

```
set.seed(1)
obs <- c(rbeta(300, 0.3, 3), rbeta(100, 4, 1.5))   # bimodal, like real data
mu <- rbeta(400, 1, 2.2); sdv <- 0.08 + 0.1 * runif(400)
sw <- dn_sweep(obs, mu, sdv)
sw          # prints a warning if the verdict reverses across thresholds
plot(sw)
```

`dn_table_export` *Export a table as CSV, Markdown or LaTeX*

Description

Export a table as CSV, Markdown or LaTeX

Usage

```
dn_table_export(x, file, digits = 3, caption = NULL)
```

Arguments

`x`	data.frame.
`file`	output path; format from extension (`.csv`, `.md`, `.tex`).
`digits`	rounding for numeric columns (default 3).
`caption`	optional caption (md/tex).

Value

invisibly, the path.

Examples

```
tf <- tempfile(fileext = ".md")
dn_table_export(data.frame(a = 1:2, b = c(0.123456, 2)), tf,
                caption = "demo")
cat(readLines(tf), sep = "\n")
```

dn_table_extremes *Extremes table (papers' Table 6 style)*

Description

The n highest and lowest regions by the area-adjusted residual, with the raw ratio shown alongside – never the ranking key.

Usage

```
dn_table_extremes(region, cddr_tab, n_obs = NULL, n_dark = NULL, n = 8)
```

Arguments

region	region names.
cddr_tab	output of dn_cddr() (with area).
n_obs, n_dark	optional richness columns to carry.
n	rows per side (default 8).

Value

data.frame, top block then bottom block.

Examples

```
set.seed(1)
area <- 10^runif(40, 4, 6.5)
x <- dn_cddr(rpois(40, 40), rpois(40, 60), area)
dn_table_extremes(paste0("eco", 1:40), x)
```

dn_weights *Coverage weight vector for one species*

Description

Per-cell weights whose inner product with a region density yields the coverage statistic: the boolean q-envelope for statistic = "q" (product = DNC_q), or 1 - dn_entryq for statistic = "auc" (product = DNC_auc). Both depend only on the density *ranking*, which is why the environmental-shift null can translate a precomputed weight array instead of re-deriving it per replicate.

Usage

```
dn_weights(Z, q = 0.95, statistic = c("auc", "q"))
```

Arguments

Z	species density (matrix or vector).
q	envelope level.
statistic	"q" or "auc".

Value

numeric vector of per-cell weights.

Examples

```
sp <- dn_space(cbind(rnorm(3000), rnorm(3000)), R = 50)
Z <- dn_density(cbind(rnorm(200), rnorm(200)), sp)
w_q   <- dn_weights(Z, statistic = "q")     # boolean envelope
w_auc <- dn_weights(Z, statistic = "auc")   # graded, threshold-free
cor(w_q, w_auc)
```

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